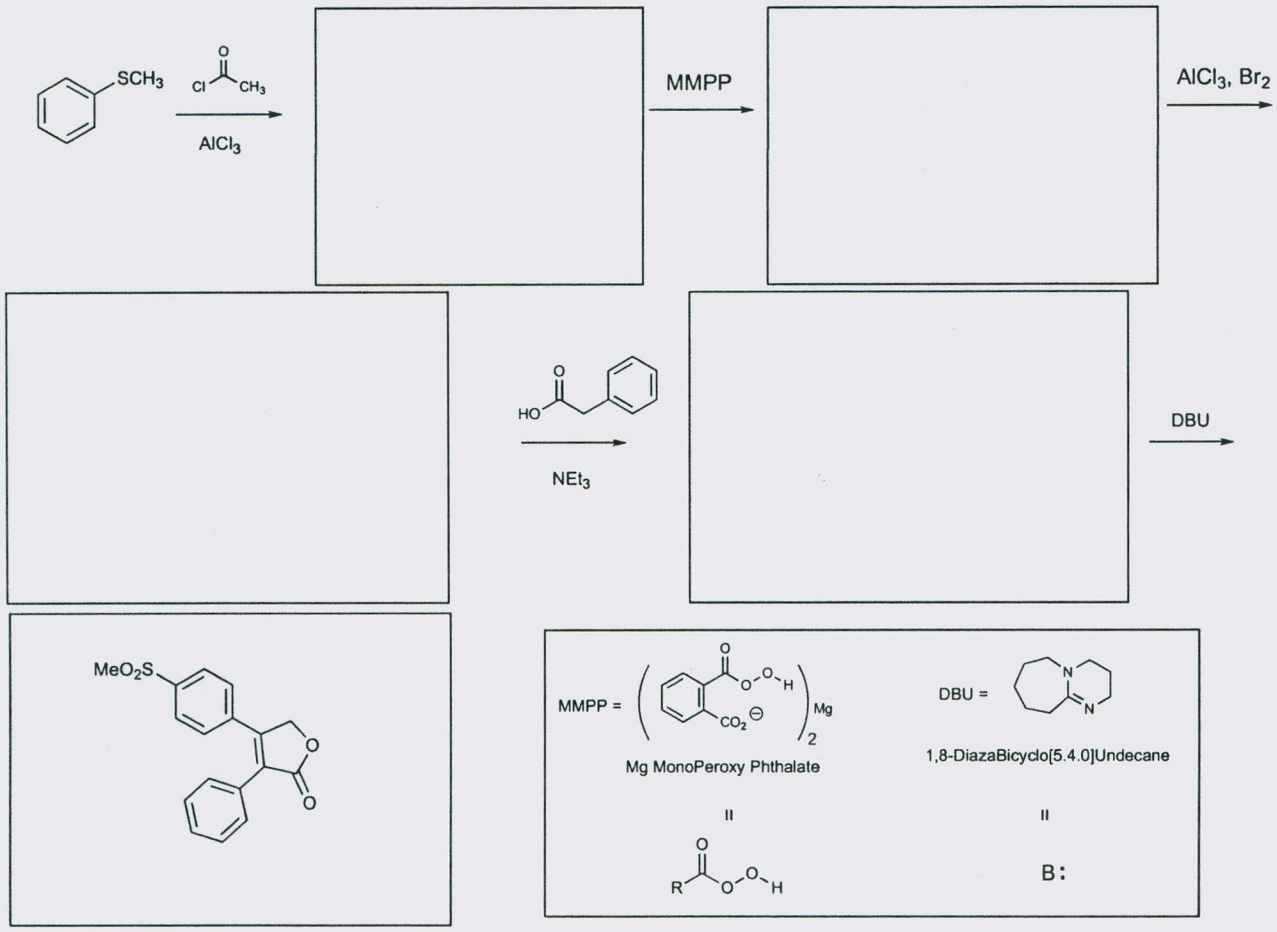
Organic Cume 03/11/2011

The following questions were taken from an exam in an undergraduate honors course on drug development. Be sure all answers are in your blue book. Donot write answers on the exam!

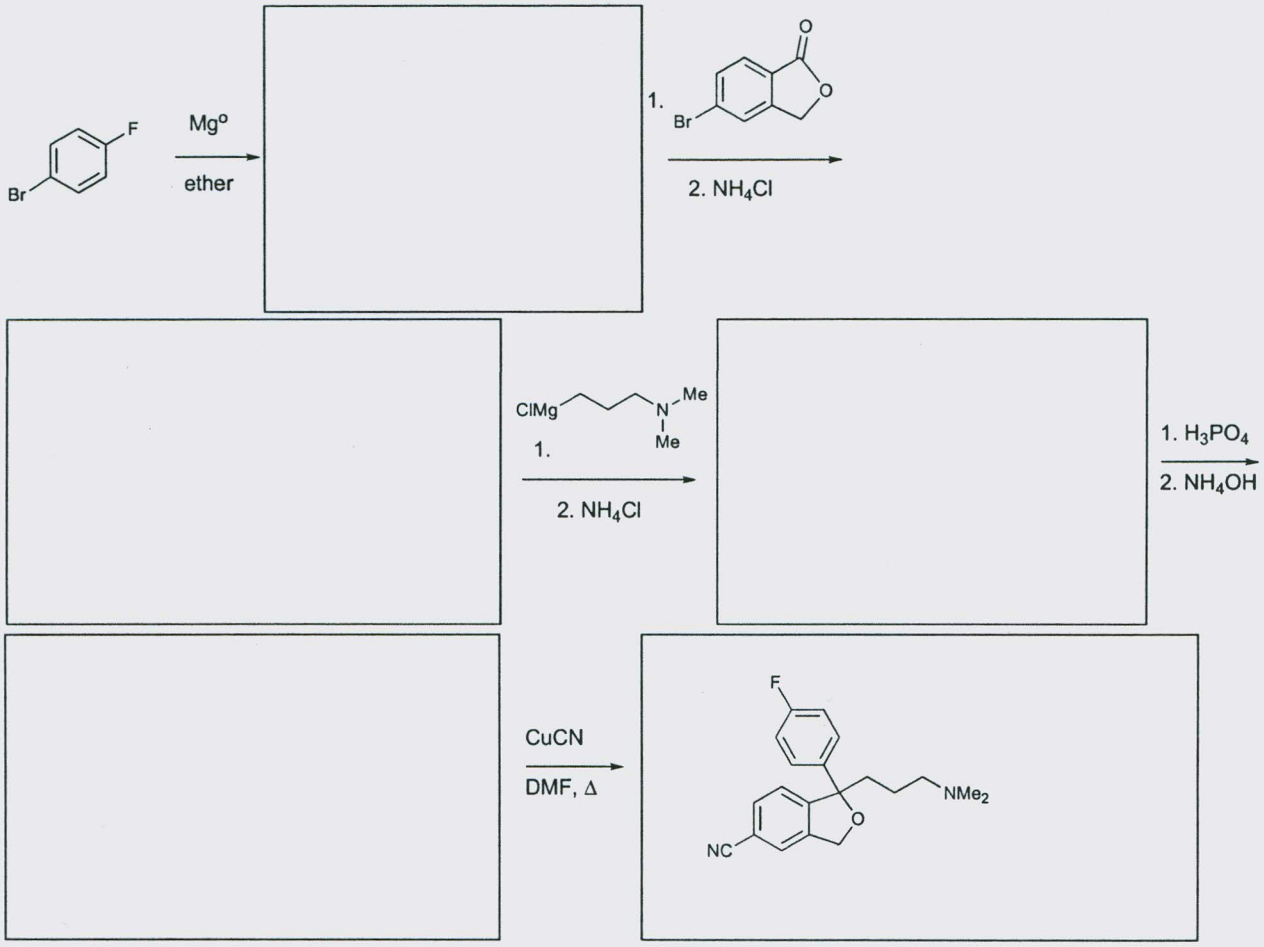
1. Draw the products of the following reaction sequence in the synthesis of the antiinflammatory COX-2 inhibitor Vioxx (rofecoxib).



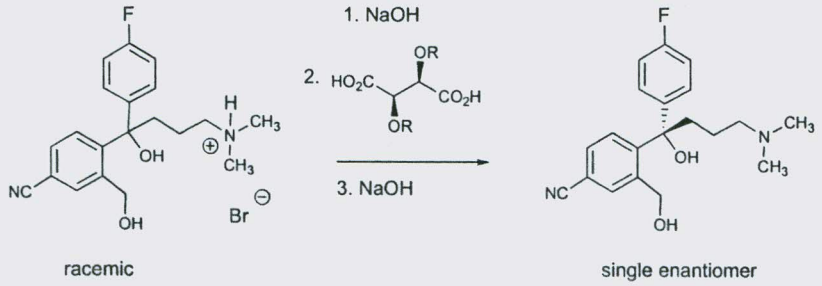
Vioxx

2. Draw the complete step-by-step electron-pushing mechanism for any step <u>except 4</u> in the above synthesis. For the purposes of the mechanism, MMPP can be abbreviated as a generic peracid (see above) and DBU as a generic base "B:".

3. Draw the products of the following reaction sequence for the synthesis of the antidepressant Celexa (citalopram).



4. In the synthesis of Lexapro (*S*-citalopram or escitalopram), briefly describe (with drawings) the process of resolution, by which a single enantiomer of the racemic intermediate below can be isolated using the enantiopure tartaric acid derivative shown. Include in your description the stereochemical and physical basis for the resolution process.



racemic

single enantiomer